

SAURABH PATEL

Electronics and Communication Engineering | National Institute of Technology Nagaland
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EDUCATION

National Institute of Technology Nagaland • Bachelor's in Electronics and Communication Engineering	Aug 2024 – June 2028
St. Xavier School, Bhiwadi • 12th (PCM)	2021 – 2022
St. Xavier School, Bhiwadi • 10th	2019 – 2020

POSITIONS OF RESPONSIBILITY

Robotics Secretary, Robotics Club, NIT Nagaland	Aug 2026 – Present
Internship Coordinator, NIT Nagaland	Aug 2026 – Present
Member, Institution's Innovation Council (IIC)	Feb 2026
Assistant Club Secretary – Robotics • Organized technical events including Clash of Wheels (RC Car Race) , Aerial Adrenaline (Drone Race) , Integration Bee , and Circuit X .	Aug 2025 – Aug 2026

WORK EXPERIENCE

Department of Telecommunications (DoT) • Sanchar Mitra • Assisted in promoting awareness of telecom services and digital communication initiatives. • Supported outreach activities and public engagement programs.	Aug 2025 – Present
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PROJECTS

Major Projects

MineSafe-X – Low-Visibility Safety & Fleet Monitoring System

ESP32, GPS, mmWave Radar, Ultrasonic, ToF, IMU, ESP-NOW

- Developed an ESP32-based V2V/V2I safety system for mining vehicles operating in fog and low-visibility conditions.
- Integrated multi-sensor distance monitoring, fleet tracking, and real-time collision warnings.

Aerial Fire Detection & Suppression Drone

STM32, ESP32, Raspberry Pi, Edge Impulse

- Built an RC drone prototype using ML-based fire detection, GPS/IMU sensing, and automated payload deployment.
- Designed the system for rapid fire-response applications.

AC Line Fault Detection System

- Developed a device for detecting and indicating fault conditions in AC electrical lines using embedded electronics and sensors.

Minor Projects

- Ultrasonic Obstacle Detection System** — Arduino-based real-time distance measurement and obstacle detection.
- Bluetooth-Controlled RC Car** — HC-05 based wireless motor control system.
- Sensor-Based Automation Projects** — Microcontroller-based projects involving sensors, actuators, and real-time monitoring.

ACHIEVEMENTS & RECOGNITION

- Invited Exhibitor – TXG Expo, Nagaland (2026)** — Invited to showcase and demonstrate an **Aerial Recognition Project**.
- Winner – Robo GP**, Technical Board, NIT Nagaland (Mar 2025)
- 1st Runner Up – Quiz Competition**, Science Society, NIT Nagaland (Feb 2025)
- 1st Runner Up – Quiz Competition**, Science Society, NIT Nagaland (Feb 2026)
- Assembly of Obstacle Avoidance Based on Multirotor UAV**, NIELIT (Sep 2025)

TECHNICAL SKILLS

Programming: C, MySQL

Microcontrollers: Arduino, ESP32, STM32

Embedded Systems: Sensor Integration, GPS, Wireless Communication, Motor Control

Robotics & UAV: RC Vehicles, Multirotor Systems, Flight Controllers

AI/ML: Edge Impulse

Tools: CorelDRAW, MS Office, Canva

LANGUAGES

Hindi, English, Haryanvi